Example 6.14

The weekly sales of two products A and B were recorded as given below

Product *A*	59	75	27	63	27	28	56
Product *B*	150	200	125	310	330	250	225

Find out which of the two shows greater fluctuations in sales.

Solution:

For comparing the fluctuations in sales of two products, we will prefer to calculate coefficient of variation for both the products.

Product A: Let $A = 56$ be the assumed mean of sales for product A.

Sales(x)	**Frequency(f)**	**A= 56** $d = x - A$	fd	fd^2
27	2	-29	-58	1682
28	1	-28	-28	784
56 A	1	0	0	0
59	1	3	3	9
63	1	7	7	49
75	1	19	19	361
Total	**7**		**-57**	**2885**

$$\bar{x} = A + \frac{\Sigma fd}{N}$$

$$= 56 - \frac{57}{7} = 47.86$$

$$\text{Variance } \sigma^2 = \frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2$$

$$= \frac{2885}{7} - \left(\frac{-57}{7}\right)^2$$

$$= 412.14 - 66.30 = 345.84$$

$$\text{Standard deviation } \sigma = \sqrt{345.84} = 18.59$$

$$\text{Coefficient of variation } (C.V) = \frac{\sigma}{x} \times 100$$

$$= \frac{18.59}{47.86} \times 100$$

$$= 38.84\ \%$$

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